

Bing Fu

CONTACT INFORMATION	83 Tat Chee Ave., Kowloon Tong Kowloon, Hong Kong, China Website: https://fulabsmimaging.github.io/fulab.github.io/	bingfu@cityu.edu.hk +852 3442 2045
EMPLOYMENT	City University of Hong Kong Assistant Professor, Biomedical Engineering, July 2023 - present	
PROFESSIONAL PREPARATION	Cornell University, Ithaca, U.S., June 2017 - June 2023 Postdoctoral Associate Advisor: Peng Chen Research: Live-cell fluorescence imaging University of Michigan, Ann Arbor, U.S. Ph.D. in Chemistry, May 2017 Advisor: Julie S. Biteen Dissertation: <i>Interactions between Fluorescent Molecules and Plasmonic Nanoparticles: A Super Resolution Study</i> Shandong University, Jinan, China B.A. in Chemistry, July 2012	
SELECT HONORS AND AWARDS	Outstanding Early Career Board Member Award, Chemical & Biomedical Imaging journal, 2026 Honorable mention, Graduate Women in Science National Fellowship Program, U.S., 2022 Karle symposium travel award, Department of Chemistry, University of Michigan, 2016 Rackham Graduate School Summer Awards, University of Michigan, 2016 Departmental fellowship, Department of Chemistry, University of Michigan, 2015-2016 Florence Fenwick Outstanding Graduate Student Instructor Award, Department of Chemistry, University of Michigan, 2015 Rackham Graduate Student Research Grant, 2014-2015 Outstanding Summer Camp Members, Peking University, 2011 Second Grade Scholarship for Excellent Students, Shandong University, 2009-2010 Outstanding Student for Academic Achievements, School of Chemistry and Chemical Engineering, Shandong University, 2010	
PUBLICATIONS	† equal contribution, * corresponding author	
	[1] Yuxi Yan, Ziqi Wu, Yuhong Sun, Miao Zhang, Bixi Zhao, Cailong Nie, Zhanwen Cheng, Qing Yang, Liming Chen, Qiqi Hao, Bing Fu , and Yu Xia*. Coupling filter mating with nanopore selective sequencing expands capturable conjugative plasmid recovery in biological wastewater treatment systems. <i>Nature Water</i> (2026).	
	[2] Chenwei Xiong†, Qiqi Hao†, Xiaoming Zhu, Chenming Lu, Yang Zhou, Xusan Yang, Feifan Wang, and Bing Fu *. Single-molecule fluorescence imaging and tracking in live bacteria: observing proteins in action. <i>Microbiology and Molecular Biology Reviews</i> 90 , e00138–25 (2026).	
	[3] Youngchan Park, Tianlei Yan, Zhiheng Zhao, Bing Fu , Muwen Yang, Farshid Salimijazi, Buz Barstow, and Peng Chen*. Spatiotemporal organization of membrane protein controls bacterial extracellular electron transfer. <i>Nature Communications</i> pages 1159–1168 (2026).	
	[4] Wenjie Li, Muwen Yang, Zhiheng Zhao, Ming Zhao, Rong Ye, Bing Fu , and Peng Chen*. Single-molecule reaction mapping uncovers diverse behaviours of electrocatalytic surface Pd-H intermediates. <i>Nature Catalysis</i> 8 (11), 1159–1168 (2025).	

- [5] Weidong Zhang[†], Chenwei Xiong[†], Peng Chen*, **Bing Fu***, and Xianwen Mao*. Elucidating Energy Conversion Pathways at Biotic/Abiotic Interfaces in Microbe-Semiconductor Hybrids. *Journal of the American Chemical Society* **147**, 20171–20188 (2025).
- [6] Mokshin Suri, Farshid S. Jazi, Jack C. Crowley, Youngchan Park, **Bing Fu**, Peng Chen, Warren R. Zipfel, Buz Barstow, and Tobias Hanrath*. Spatially resolved charge-transfer kinetics at the quantum dot-microbe interface using fluorescence lifetime imaging microscopy. *Proceedings of the National Academy of Sciences* **122**(12), e2407987122 (2025).
- [7] Wenyao Zhang, Christine E. Harper, Junsung Lee, **Bing Fu**, Malissa Ramsukh, Christopher J. Hernandez, and Peng Chen*. Transporter excess and clustering facilitate adaptor-protein shuttling for bacterial efflux. *Cell Reports Physical Science* **6**, 102441 (2025).
- [8] Shuchi Zhang, Deqi Fan, Qingdian Yan, Yi Lu, Donglei Wu, **Bing Fu**, and Ming Zhao*. Single-molecule fluorescence imaging of photocatalytic nanomaterials. *Journal of Materials Chemistry A* **12**, 19627–19662 (2024).
- [9] **Bing Fu**[†], Xianwen Mao[†], Youngchan Park, Zhiheng Zhao, Tianlei Yan, Won Jung, Danny Duckworth, Wenjie Li, Brooke Pian, Farshid Salimijazi, Mokshin Suri, Tobias Hanrath, Buz Barstow, and Peng Chen*. Single-cell multimodal imaging uncovers energy conversion pathways in biohybrids. *Nature Chemistry* **15**, 1400–1407 (2023).
- [10] **Bing Fu**[†], Kushal Sengupta[†], Lauren A Genova[†], Ace George Santiago, Won Jung, Lukasz Krzemiński, Udit Kumar Chakraborty, Wenyao Zhang, and Peng Chen*. Metal-induced sensor mobilization turns on affinity to activate regulator for metal detoxification in live bacteria. *Proceedings of the National Academy of Sciences* **117**(24), 13248–13255 (2020).
- [11] Feifan Wang and **Bing Fu**. 4 - Anion-exchange membranes for direct methanol alkaline fuel cells. In Kingshuk Dutta, editor, *Direct Methanol Fuel Cell Technology*, pages 71–106. Elsevier, (2020).
- [12] Lauren A Genova[†], Melanie F Roberts[†], Yu-Chern Wong, Christine E Harper, Ace George Santiago, **Bing Fu**, Abhishek Srivastava, Won Jung, Lucy M Wang, Lukasz Krzemiński, Xianwen Mao, Xuanhao Sun, Chung-Yuen Hui, Peng Chen*, and Christopher J Hernandez*. Mechanical stress compromises multicomponent efflux complexes in bacteria. *Proceedings of the National Academy of Sciences* **116**(51), 25462–25467 (2019).
- [13] **Bing Fu**, Benjamin P Isaacoff, and Julie S Biteen*. Super-Resolving the Actual Position of Single Fluorescent Molecules Coupled to a Plasmonic Nanoantenna. *ACS Nano* **11**(9), 8978–8987 (2017).
- [14] **Bing Fu**, Jessica D Flynn, Benjamin P Isaacoff, David J Rowland, and Julie S Biteen*. Super-Resolving the Distance-Dependent Plasmon-Enhanced Fluorescence of Single Dye and Fluorescent Protein Molecules. *The Journal of Physical Chemistry C* **119**(33), 19350–19358 (2015).
- [15] Esther Wertz, Jessica D Flynn, Courtney N Talicska, **Bing Fu**, Benjamin P Isaacoff, and Julie S Biteen*. Plasmon-enhanced fluorescence permits super-resolution imaging in vitro and in cells. *Laser Science* page LTh3H.6 (2013).

PRESENTATIONS

“Synergistic Chemotaxis and Metal Regulation Enable Pathogen Mobility at the Host–Pathogen Interface”, Invited speaker, The 32nd Federation of Asian and Oceanian Biochemists and Molecular Biologists (FAOBMB) Conference, Hong Kong, Aug. 2026

“Live-cell single-molecule imaging and tracking reveals novel mechanisms in bacterial stress response”, Short talk, Microbial Stress Response Gordon Research Conference, Mount Holyoke, July 2026

“Single-molecule imaging of signal transduction in microbes”, Invited Speaker, 2nd International Symposium on Chemical and Biomedical Imaging (CBMI-II), Hong Kong, May 2026

“Single-cell single-molecule super-resolution imaging and tracking in live cells”, Invited talk, 2nd BMEMat Youth Editorial Board meeting, Jinan, Jan. 2026

“Super-resolution single-molecule fluorescence imaging reveals protein mechanisms in microbes”, Keynote speaker, Annual Academic Meeting of Chinese Society for Microbiology, Hangzhou, China, Oct. 2025

“High-resolution fluorescence imaging in microbial and mammalian cells”, Invited talk, 2nd BMEMat-Advanced Materials Joint Symposium, Jinan, July 2025

“Watching molecules one at a time: high-resolution single-molecule fluorescence imaging in microbial and mammalian cells”, Invited talk, CityU-NUS Joint Symposium, CityUHK, Hong Kong, April 2025

“Single-cell single-molecule fluorescence imaging in bacteria for energy and health”, Invited talk, SIAT Chinese Academy of Sciences, Shenzhen, Nov. 2024

“Single-cell single-molecule fluorescence imaging in bacteria for energy and health”, Invited talk, HKUST, Hong Kong, Sept. 2024

“Single-cell imaging in nano-bio hybrids for studying energy conversion”, Keynote Speaker, 1st International Symposium on Chemical and Biomedical Imaging (CBMI-I), Nanjing, Aug. 2024

“Super-resolution fluorescence imaging in live microbial systems”, iCANX Youth Talks, online, July 2024

“Seeing bacteria beyond diffraction limit: Application of single-molecule super-resolution imaging in studying molecular mechanisms in microbes”, Invited talk, SusTech, School of Environmental Science and Engineering, Shenzhen, March 2024

“Electron Transfer Mechanism at the Microbe-Semiconductor Interface with Single-cell Resolution”, Invited talk, The 2024 International Conference on Bacteria and Global Health, Shenzhen, Jan. 2024

“Single-molecule study of copper regulation two-component system CusS-CusR in *E. coli*”, Best Poster Award, Annual Academic Meeting of Chinese Society for Microbiology, Shanghai, China, Oct. 2023

“Dissecting the signal transduction dynamics for copper homeostasis in live *E. coli*”, Invited talk, American Chemical Society (ACS) National Meeting and Exposition, Chicago, IL, USA, Aug. 2022

“Proteins are smart: A single-molecule study of metal homeostasis signal transduction in *E. coli*”, Invited talk, University of Chicago Future Faculty Conference, Chicago, IL, USA, May 2022

“Proteins are smart - a single molecule study of CusRS system for efficient Cu homeostasis in *E. coli*”, Oral presentation, 5th Annual Cornell Biophysics Symposium, Ithaca, NY, USA, Nov. 2019

“Single Molecule super-resolution study of distance-dependent plasmonic fluorescence enhancement and emission mislocalization”, Oral presentation, 253rd American Chemical Society (ACS) National Meeting and Exposition, San Francisco, CA, USA, Apr. 2017

“Single-molecule super-resolution microscopy study of the distance-dependent interaction between a fluorescent molecule and a nano-antenna”, Oral presentation, 252nd American Chemical Society (ACS) National Meeting and Exposition, Philadelphia, PA, USA, Aug. 2016

“Study of the Distance-Dependent Interaction between Fluorescent Molecule and Nano-Antennas by Single-Molecule Super-Resolution Microscopy”, Poster presentation, Gordon Research Conference, South Hadley, MA, USA, June, 2016

“Single-molecule super-resolution microscopy study of the distance-dependent interaction between a fluorescent molecule and a nano-antenna”, Oral presentation, 47th ACS Central Regional Meeting, Covington, KY, USA, May 2016

“Super-Resolution Study of the Distance Dependence of Single-Molecule Plasmon-Enhanced Fluorescence”, Oral presentation, Materials Research Society (MRS) Spring Meeting and Exhibit, San Francisco, CA, USA, April 2015

“Distance-Dependent Plasmon-Induced Fluorescence Enhancement of Single Molecules on a Gold Nanoparticle Substrate”, Oral presentation, Victor C. Vaughan Symposium, Ann Arbor, MI, USA, Aug. 2014

“Distance-Dependent Plasmon-Induced Fluorescence Enhancement of Single Molecules on a Gold Nanoparticle Substrate”, Poster presentation, Gordon Research Conference, South Hadley, MA, USA, June, 2014

EXTERNAL GRANT RGC General Research Fund, 2027-ongoing, Co-I
Deciphering the Role of Methyltransferase M. Fcohsdm in Flavobacterium columnare Virulence and Its Application as a Novel Live Attenuated Vaccine

RGC Collaborative Research Equipment Grant, 2026-ongoing, Co-PI
The First State-of-the-Art Molecular Resolution Light Microscope for Live Cell Imaging into our Advanced Imaging Platforms to Timely Promote Interdisciplinary and Advanced Life Sciences Research in Hong Kong and Beyond

RGC Early Career Scheme, 2025-ongoing, PI
Interplay between Microbial Metal Regulation and Chemotaxis at the Host-Pathogen Interface Studied by Single-molecule Super-resolution Fluorescence Microscopy

NSFC Young Scientist Fund, 2025-ongoing, PI
Live-cell Exploration of Cpxa/Cpxr Dynamics in Signal Transduction for Antibiotic Resistance by Super-Resolution Single-Molecule Imaging

INTERNAL GRANT TDG (CityUHK), 2026-ongoing, PI
From structure to solution: A team-based, student-driven approach to drug discovery

TEACHING Department of Biomedical Engineering, CityU Hong Kong
Course leader: BME4104 Technology for Drug Discovery, 2023-present
Course leader: BME6414 Fundamentals and Applications of Single-molecule Biophysics in Rapid Diagnostics, 2025-present

Department of Chemistry, University of Michigan
Graduate Student Instructor
CHEM 125 - General Chemistry lab, Sept. 2012 - Dec. 2013
CHEM 130 - General Chemistry, June 2013 - Aug. 2013
CHEM 125 - General Chemistry lab, June 2014 - Aug. 2014
Graduate Student Administrator
CHEM 125 - General Chemistry lab, Sept. 2014 - Dec. 2015

SERVICE AND OUTREACH
Chemical & Biomedical Imaging, Early Career Advisory Board Member, 2023-present
BMEMat, Youth Editorial Board Member, 2023-present
Graduate Women in Science Membership, 2021-2022
Discussion leader, Responsible Conduct of Research Symposium, Cornell University, 2019
American Chemical Society Member, 2016-2017
Sponsored AAAS/Science Membership, nominated by Department Chair, 2015-2017
Poster judge at Vaughan Symposium, University of Michigan, 2014
Scientific Content Reviewer, reviewed more than 30 publications in journals including:
Nature Methods
ACS Photonics
ACS Sensors
ACS Nano
Journal of the American Chemistry Society

